Homework 6

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Physics 15
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Question 1. Find the de Broglie wavelength of (a) an electron with kinetic energy of 1 eV; (b) an electron with kinetic energy of 10 MeV; (c) a neutron with kinetic energy of 10 MeV; (d) a 500 g grapefruit that has been thrown with a speed of 10 m/s. Note: make sure to consider whether you need to use relativistically correct equations in each case!

Solution. The de Broglie wavelength is given by $\lambda = \frac{h}{p}$. We must determine whether to use classical or relativistic momentum based on the ratio of kinetic energy K to rest mass energy m_0c^2.

For the slower electron, since $K = 1$ eV $\ll mc^2 = 0.511$ MeV, we use non-relativistic kinematics. This implies that $K = \frac{p^2}{2m}$. This can be rearranged such that $p = \sqrt{2mK}$, and hence

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{\sqrt{2(9.11 \times 10^{-31}\ \text{kg})(1.602 \times 10^{-19}\ \text{J})}} \approx 1.23 \times 10^{-9}\ \text{m} = \boxed{1.23\ \text{nm.}}$$

For the faster electron, since $K = 10$ MeV $\gg mc^2 = 0.511$ MeV, we must use relativistic kinematics. It is true that

$$E^2 = (\gamma mc^2)^2 = (pc)^2 + (mc^2)^2.$$

If we add $-2(\gamma mc^2)(mc^2) + (mc^2)^2$ to both sides, we get

$$\begin{aligned}(\gamma mc^2)^2 - 2(\gamma mc^2)(mc^2) + (mc^2)^2 &= (pc)^2 + (mc^2)^2 - 2(\gamma mc^2)(mc^2) + (mc^2)^2. \\ (\gamma mc^2 - mc^2)^2 &= (pc)^2 - 2mc^2(\gamma mc^2 - mc^2) \\ K^2 &= (pc)^2 - 2Kmc^2 \implies pc = \sqrt{K^2 + 2Kmc^2}.\end{aligned}$$

Therefore, this can be substituted such that

$$\begin{aligned}\lambda &= \frac{hc}{\sqrt{K^2 + 2Kmc^2}} \\ &= \frac{1240\ \text{MeV} \cdot \text{fm}}{\sqrt{(10\ \text{MeV})^2 + 2(10\ \text{MeV})(0.511\ \text{MeV})}} \approx \frac{1240}{10.5}\ \text{fm} \approx \boxed{118\ \text{fm.}}\end{aligned}$$

For the neutron, $mc^2 \approx 939.6$ MeV. Since $K = 10$ MeV $\ll 939.6$ MeV, the non-relativistic formula is appropriate:

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{6.626 \times 10^{-34} \text{ J s}}{\sqrt{2(1.675 \times 10^{-27} \text{ kg})(10 \text{ MeV} \times \frac{10^6 \text{ eV}}{\text{MeV}} \times \frac{1.602 \times 10^{-19} \text{ J}}{\text{eV}})}}$$
$$\approx 9.04 \times 10^{-15} \text{ m} = \boxed{9.04 \text{ fm.}}$$

For the grapefruit, we use the standard formula $p = mv$ so that

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(0.5 \text{ kg})(10 \text{ m/s})} = \boxed{1.33 \times 10^{-34} \text{ m.}}$$

$\square$

Question 2. Uncertainty Principle: Complete Eisberg and Resnick: 3.29 and 3.34

29. The lifetime of an excited state of a nucleus is usually about 10^{-12} sec. What is the uncertainty in energy of the γ-ray photon emitted?

Solution. Let us use the derived uncertainty relationship

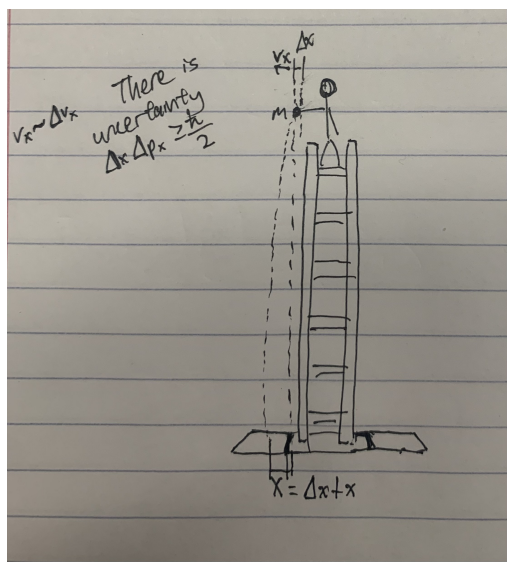
$$\Delta E \Delta t \geq \frac{\hbar}{2}.$$

Minimizing and rearranging the equation gives us

$$\Delta E = \frac{\hbar}{2\Delta t} \approx \frac{6.626 \times 10^{-34} \text{ J s}/2\pi}{2(10^{-12} \text{ s})} \approx \boxed{5.27 \times 10^{-23} \text{ J.}}$$

□

34. A boy on top of a ladder of height H is dropping marbles of mass m to the floor and trying to hit a crack in the floor. To aim, he is using equipment of the highest possible precision. (a) Show that the marbles will miss the crack by an average distance of the order of $(\hbar/m)^{1/2}(H/g)^{1/4}$, where g is the acceleration due to gravity. (b) Using reasonable values of H and m, evaluate this distance.



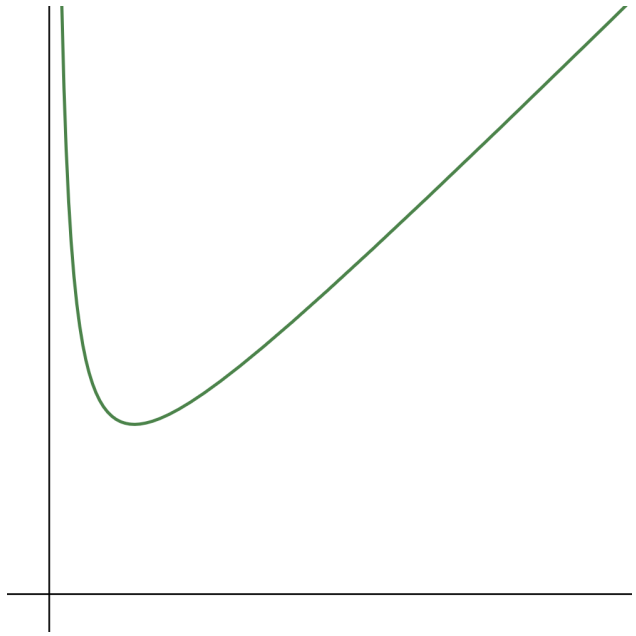
Solution. Let us first find v_x. Using basic kinematics and minimizing the uncertainty relationship $\Delta x \Delta p_x = \Delta x m \Delta v_x \geq \frac{\hbar}{2}$, we get that

$$v_x \sim \Delta v_x = \frac{\hbar}{2m\Delta x}.$$

Additionally, we must find the total time it takes for the ball to land. Since $\frac{1}{2}gt^2 = H$, we can rearrange this equality to get $t = \sqrt{\frac{2H}{g}} \sim \sqrt{\frac{H}{g}}$. This implies that the total displacement ΔX is

$$\Delta X = \Delta x + \Delta v_x t \sim \Delta x + \frac{\hbar}{m}\sqrt{\frac{H}{g}}\left(\frac{1}{\Delta x}\right)$$

Let us plot this equation where ΔX is on the y-axis and Δx is on the x-axis: To minimize



uncertainty, we need to find the smallest ΔX. This can be done by taking the derivative with respect to Δx and set it to zero:

$$\frac{d(\Delta X)}{d(\Delta x)} \sim 1 - \frac{\hbar}{m\Delta x^2}\sqrt{\frac{H}{g}} = 0$$

Then, the optimal distance Δx is

$$\Delta x \sim \left(\frac{\hbar}{m}\right)^{1/2}\left(\frac{H}{g}\right)^{1/4}.$$

Now, substituting this optimal Δx back into the expression for ΔX:

$$\begin{aligned}\Delta X &\sim \left[\left(\frac{\hbar}{m}\right)^{1/2}\left(\frac{H}{g}\right)^{1/4}\right] + \frac{\hbar}{m}\sqrt{\frac{H}{g}}\left[\left(\frac{\hbar}{m}\right)^{-1/2}\left(\frac{H}{g}\right)^{-1/4}\right] \\ &\sim \left(\frac{\hbar}{m}\right)^{1/2}\left(\frac{H}{g}\right)^{1/4} + \left(\frac{\hbar}{m}\right)^{1/2}\left(\frac{H}{g}\right)^{1/4} \\ &\boxed{\sim \left(\frac{\hbar}{m}\right)^{1/2}\left(\frac{H}{g}\right)^{1/4}.}\end{aligned}$$

In this scenario, let us choose $H = 5$ m and $m = .005$ kg. Then,

$$\Delta X \sim \left(\frac{6.626/2\pi \times 10^{-34}\ \text{J s}}{0.005\ \text{kg}}\right)^{1/2}\left(\frac{5\ \text{m}}{10\ \text{m/s}^2}\right)^{1/4} \boxed{\approx 1.22 \times 10^{-16}\ \text{m.}}$$

$\square$

Question 3. A time-independent wave function has the form of a Gaussian

$$\psi(x) = Ce^{-x^2/2a^2}$$

Find the normalization constant, C. Hint: You might want to consult an integral table for this one. It is possible to solve the integral analytically, but you don't have to!

Solution. To normalize the wavefunction, we must find C in ψ such that

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

Let us first substitute the wavefunction into the integral so that

$$|C|^2 \int_{-\infty}^{\infty} e^{-2x^2/2a^2} dx = |C|^2 \int_{-\infty}^{\infty} e^{-x^2/a^2} dx = 1.$$

Now, let us do a u-substitution. Let $u = x/a$. Then, $du = dx/a$. Substituting $dx = adu$ into the integral and adjusting the bounds (though the same) gives us

$$a|C|^2 \int_{-\infty}^{\infty} e^{-u^2} du = 1.$$

It is true that a standard Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. Therefore,

$$a|C|^2\sqrt{\pi} = 1.$$

Rearranging this in terms of C gives us $\boxed{C = \frac{1}{a^{1/2}\pi^{1/4}}.}$ □

Question 4. The aurora is caused when electrons or protons enter the atmosphere along Earth's magnetic field lines, and then collide with atmospheric atoms. An atom can be knocked into an excited state, and when it relaxes to a lower energy state, a photon is emitted. For example, the common green aurora is produced by excited oxygen. In the case of proton aurora, a proton enters the atmosphere and it can steal an electron, turning it into a neutral hydrogen atom in an excited state ($n = 4$). When this atom relaxes to a lower energy state ($n = 2$), an H-beta photon is emitted. What color is the H-beta proton aurora?

Solution. Given the Balmer equation

$$\lambda = \left[R_H \left(\frac{1}{n_f^2} - \frac{1}{n_0^2} \right) \right]^{-1},$$

where $R_H = 1.097 \times 10^7 \text{ m}^{-1}$. Since the relaxes from a $n_i = 4$ to a $n_f = 2$ state, we get that

$$\lambda = \left[1.097 \times 10^7 \text{ m}^{-1} \left(\frac{1}{2^2} - \frac{1}{4^2} \right) \right]^{-1} \approx 4.86 \times 10^{-7} \text{ m} = \boxed{486 \text{ nm.}}$$

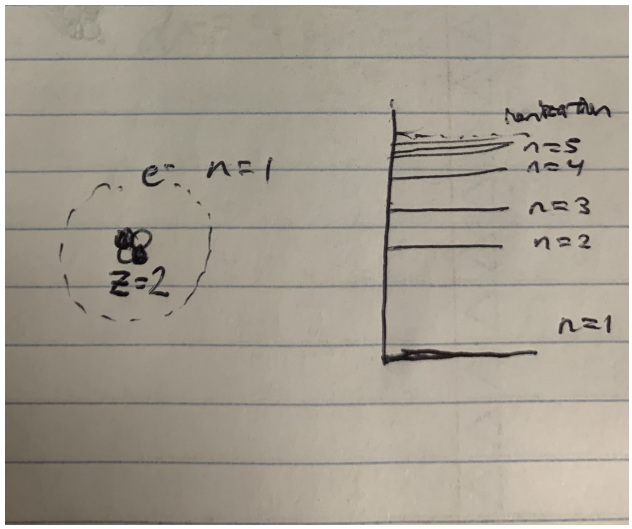
The wavelength of 486 nm corresponds to a $\boxed{\text{cyan color.}}$ □

Question 5. Helium Energy Levels

(a) Construct an energy-level diagram for the He^+ ion, for which $Z = 2$.

(b) What is the ionization energy for He^+?

Solution. For part (a) He^+ atom has two protons, two neutrons, and one electron since the +1 indicates one electron removed. Therefore, we get a energy-level diagram as such:



For part (b), the ionization energy is the energy required to move an electron from $n = 1$ to $n = \infty$ is

$$\Delta E = -13.6 \text{ eV}(Z^2)\left(\frac{1}{n_f^2} - \frac{1}{n_0^2}\right) = -13.6 \text{ eV}(2^2)\left(0 - \frac{1}{1^2}\right) = \boxed{54.4 \text{ eV.}}$$

□